

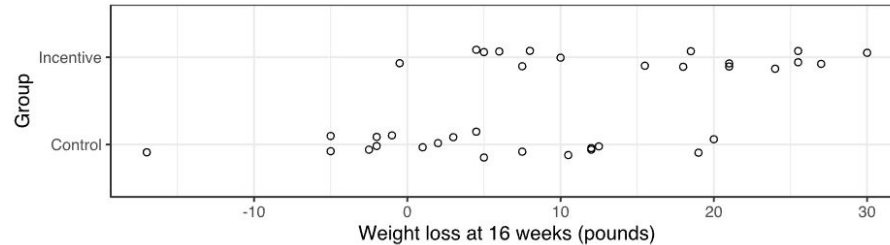
Analyzing an Experiment: Difference of Means

MATH 213

Group 1

The table and graph below show the actual experimental data from the weight-loss experiment that we studied earlier in the semester. Remember, the incentive group was given a financial incentive to lose weight.

Group	n	group mean weight loss at 16 weeks (pounds)	group std. dev. weight loss (pounds)
Control	19	3.92	9.11
Incentive	17	15.68	9.41



What parameter are we interested in estimating using the study data?

The difference in **average** weight loss between different incentive groups

What is the “point estimate” (sample statistic) for this parameter?

$$15.68 - 3.92 = 11.76$$

What is the estimated standard error for the parameter. Include some of your work.

$$SE_{\text{con}} = 9.11 / \sqrt{19} = 2.09$$

$$SE_{\text{inc}} = 9.41 / \sqrt{17} = 2.28$$

$$SE_{\text{diff}} = \sqrt{(SE_{\text{con}}^2 + SE_{\text{inc}}^2)} = 3.09$$

Use your point estimate and standard error to create and interpret a 95% confidence interval for the parameter.

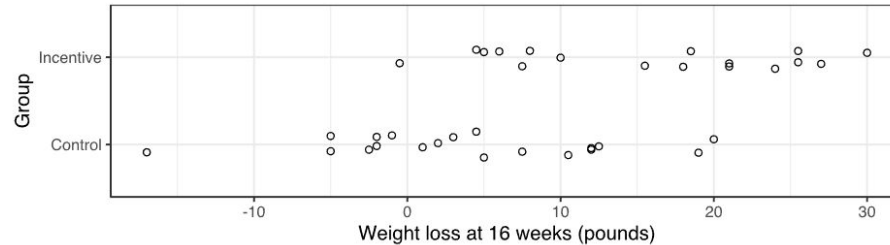
$$11.76 \pm 3.09 * 1.96 = (5.704, 17.816)$$

We are 95% confident that the average person in the incentive group will lose between 5.704 lbs and 17.816 lbs more than the average person in the control group.

Group 2

The table and graph below show the actual experimental data from the weight-loss experiment that we studied earlier in the semester. Remember, the incentive group was given a financial incentive to lose weight.

Group	n	group mean weight loss at 16 weeks (pounds)	group std. dev. weight loss (pounds)
Control	19	3.92	9.11
Incentive	17	15.68	9.41



What parameter are we interested in estimating using the study data? The difference in weight loss between financial incentives or no incentive

What is the “point estimate” (sample statistic) for this parameter? $15.68 - 3.92 = 11.76$

What is the estimated standard error for the parameter. Include some of your work.

$$\text{sqrt}((9.41/17) + (9.11/19)) = 1.01636759884$$

conf

$$11.76 + 2.0327351976 = 13.7927351977$$

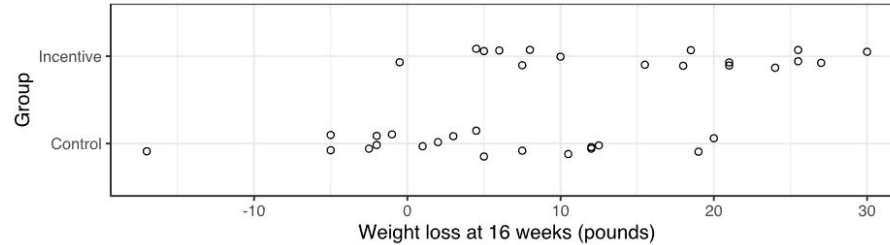
$$11.76 - 2.0327351976 = 9.72726480232$$

Use your point estimate and standard error to create and interpret a 95% confidence interval for the parameter. We are 95% confident that the true difference in mean weight loss between people who have a financial incentive and those who don't is between 9.727 and 13.793. That is, the average person planning to lose weight is likely going to lose on average between 9.727 and 13.793.

Group 3

The table and graph below show the actual experimental data from the weight-loss experiment that we studied earlier in the semester. Remember, the incentive group was given a financial incentive to lose weight.

Group	n	group mean weight loss at 16 weeks (pounds)	group std. dev. weight loss (pounds)
Control	19	3.92	9.11
Incentive	17	15.68	9.41



What parameter are we interested in estimating using the study data?

The difference in the mean weight loss between the two groups.

What is the “point estimate” (sample statistic) for this parameter?

$$15.68 - 3.92 = 11.76$$

What is the estimated standard error for the parameter. Include some of your work.

$$SE_I = 9.41/\sqrt{17} = 2.282$$

$$SE_C = 9.11/\sqrt{19} = 2.090$$

$$SE = \sqrt{(SE_I)^2 + (SE_C)^2} = \sqrt{(2.282)^2 + (2.090)^2} = 3.094$$

Use your point estimate and standard error to create and interpret a 95% confidence interval for the parameter.

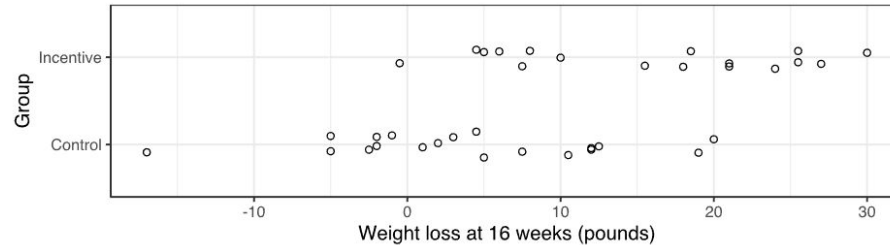
$$11.76 \pm 2*3.094 = (5.572, 17.948)$$

I am 95% confident that the difference in average weight loss between the two study groups is between 5.572 and 17.948 pounds, with the incentive group losing more weight on average than the control group.

Group 4

The table and graph below show the actual experimental data from the weight-loss experiment that we studied earlier in the semester. Remember, the incentive group was given a financial incentive to lose weight.

Group	n	group mean weight loss at 16 weeks (pounds)	group std. dev. weight loss (pounds)
Control	19	3.92	9.11
Incentive	17	15.68	9.41



What parameter are we interested in estimating using the study data?

How much more weight, on average, people would lose on the incentive plan (if all participants were on the incentive plan) than on the control plan (if all participants were on the control plan).

What is the “point estimate” (sample statistic) for this parameter?

$$15.68 - 3.92 = 11.76$$

What is the estimated standard error for the parameter. Include some of your work.

$$SE_t = 2.28$$

$$SE_c = 2.09$$

$$Geomean = ((SE_t)^2 + (SE_c)^2)^{.5} = 3.09$$

Use your point estimate and standard error to create and interpret a 95% confidence interval for the parameter.

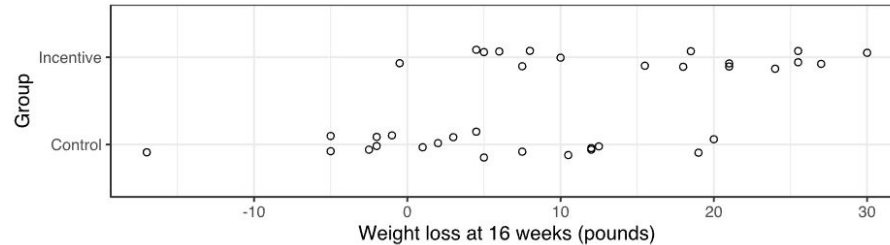
$$11.79 \pm (2 \times 3.09) = (5.58, 17.94)$$

We are 95% confident that if everyone was on the incentive plan they would lose somewhere between 5.58lbs and 17.94lbs more than if everyone was in the control group with no incentive plan.

Group 5

The table and graph below show the actual experimental data from the weight-loss experiment that we studied earlier in the semester. Remember, the incentive group was given a financial incentive to lose weight.

Group	n	group mean weight loss at 16 weeks (pounds)	group std. dev. weight loss (pounds)
Control	19	3.92	9.11
Incentive	17	15.68	9.41



What parameter are we interested in estimating using the study data?

The difference in weight loss amongst different incentive groups

What is the “point estimate” (sample statistic) for this parameter?

$$15.68 - 3.92 = 11.76$$

What is the estimated standard error for the parameter. Include some of your work.

$$SE_c = 9.11 / \sqrt{19} = 2.08$$

$$SE_i = 9.41 / \sqrt{17} = 2.28$$

$$\sqrt{2.08^2 + 2.28^2} = 3.08$$

Use your point estimate and standard error to create and interpret a 95% confidence interval for the parameter.

Sample statistic $\pm t^*(SE)$

$$11.76 + (2 * 3.08) = 17.92$$

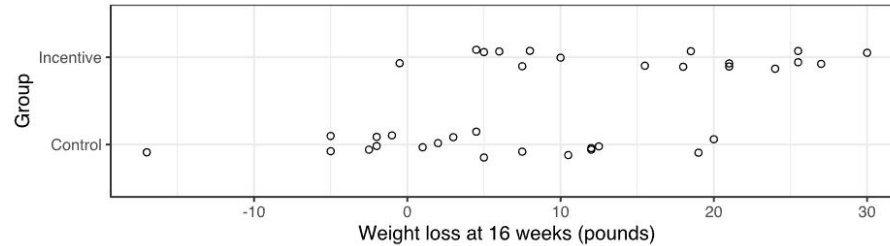
$$11.76 - (2 * 3.08) = 5.6$$

We are 95% confident that the proportion of average weight loss among the incentive group is between 5.6 to 17.92 higher than the average weight loss of the control group.

Group 6

The table and graph below show the actual experimental data from the weight-loss experiment that we studied earlier in the semester. Remember, the incentive group was given a financial incentive to lose weight.

Group	n	group mean weight loss at 16 weeks (pounds)	group std. dev. weight loss (pounds)
Control	19	3.92	9.11
Incentive	17	15.68	9.41



What parameter are we interested in estimating using the study data?

Difference in weight loss means at 16 weeks between the control group and those given a financial incentive

What is the “point estimate” (sample statistic) for this parameter?

$$3.92 - 15.68 = \underline{-11.76}$$

What is the estimated standard error for the parameter. Include some of your work.

$$\text{SE of control: } 9.11 / \sqrt{19} = 2.09$$

$$\text{SE of incentive: } 9.41 / \sqrt{17} = 2.2$$

$$\text{SE of difference between means: } \sqrt{2.09^2 + 2.2^2} = 3.03$$

Use your point estimate and standard error to create and interpret a 95% confidence interval for the parameter.

$$\text{Lower bound: } -11.76 - (2 * 3.03) = -17.82$$

$$\text{Upper bound: } -11.76 + (2 * 3.03) = -5.7$$

$$95\% \text{ CI: } (-17.82, -5.7)$$

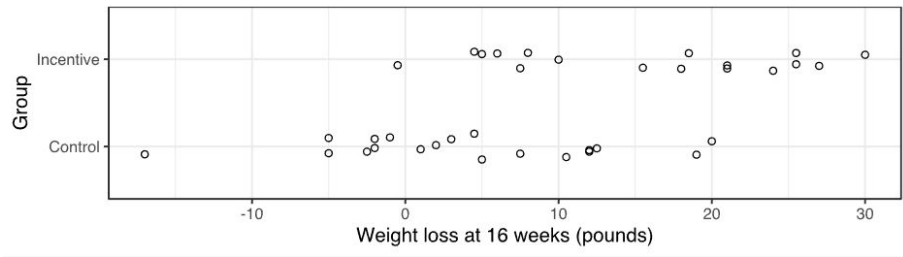
We are 95% confident that the population's difference in mean weight loss at 16 weeks between the control group and the financial incentive group is between -17.82 and -5.7.

We are 95% confident that the participants without an incentive will have between 5.7 and 17.82 pounds lower weight loss compared to participants given a financial incentive.

Group 7

The table and graph below show the actual experimental data from the weight-loss experiment that we studied earlier in the semester. Remember, the incentive group was given a financial incentive to lose weight.

Group	n	group mean weight loss at 16 weeks (pounds)	group std. dev. weight loss (pounds)
Control	19	3.92	9.11
Incentive	17	15.68	9.41



What parameter are we interested in estimating using the study data?
What is the difference in mean weight lost between the group with no financial incentive and one with financial incentive

What is the “point estimate” (sample statistic) for this parameter?

Difference in means

11.76

What is the estimated standard error for the parameter. Include some of your work.

$\sigma/\sqrt{n}$

Control: $SE_{diff} = \sqrt{(se1)^2 + (se2)^2}$

$9.11/\sqrt{19}$ $SE_{diff} = 3.09$

2.09

Incentive:

$9.41/\sqrt{17}$

2.28

Use your point estimate and standard error to create and interpret a 95% confidence interval for the parameter.

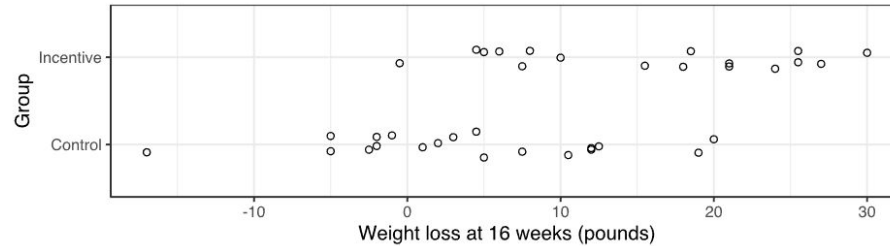
(5.54, 17.9)

We are 95% confident that the true difference in mean weight lost between the group with financial incentive and the group without financial incentive is between 5.54 and 17.9 lbs.

Group 8

The table and graph below show the actual experimental data from the weight-loss experiment that we studied earlier in the semester. Remember, the incentive group was given a financial incentive to lose weight.

Group	n	group mean weight loss at 16 weeks (pounds)	group std. dev. weight loss (pounds)
Control	19	3.92	9.11
Incentive	17	15.68	9.41



What parameter are we interested in estimating using the study data?
The difference in loss of weight between the control and incentive group.

What is the “point estimate” (sample statistic) for this parameter?

$$15.68 - 3.92 = 11.76$$

What is the estimated standard error for the parameter. Include some of your work.

$$SE_{\text{control}} = o / \sqrt{n}$$

$$SE_{\text{control}} = 9.11 / \sqrt{19}$$

$$SE_{\text{control}} = 2.08997733556$$

$$SE_{\text{incentive}} = 9.41 / \sqrt{17}$$

$$SE_{\text{incentive}} = 2.28226023159$$

$$SE = \sqrt{((2.08997733556)^2) + ((2.28226023159)^2)}$$

$$SE = 3.09462712259$$

Use your point estimate and standard error to create and interpret a 95% confidence interval for the parameter.

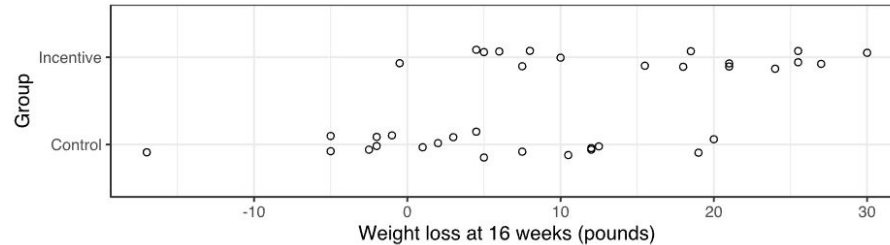
$$11.76 \pm (1.96 * 3.09462712259) = (5.69, 17.83)$$

We are 95% confident that the proportion of average weight loss between both the control and incentive groups is between 5.69 lbs and 17.83 lbs.

Group 9

The table and graph below show the actual experimental data from the weight-loss experiment that we studied earlier in the semester. Remember, the incentive group was given a financial incentive to lose weight.

Group	n	group mean weight loss at 16 weeks (pounds)	group std. dev. weight loss (pounds)
Control	19	3.92	9.11
Incentive	17	15.68	9.41



What parameter are we interested in estimating using the study data?

We are trying to find the difference in mean weight loss between the two incentive groups.

What is the “point estimate” (sample statistic) for this parameter?

$$15.68 - 3.92 = 11.76$$

The point estimate is the difference in means for the parameter.

What is the estimated standard error for the parameter. Include some of your work.

Sample SD / sqrt(n)

$$SE_{\text{control}} = 9.11/\text{sqrt}(19) = 2.08$$

$$SE_{\text{incentive}} = 9.41/\text{sqrt}(17) = 2.28$$

$$SE = \text{sqrt}(2.08^2 + 2.28^2) = 3.08$$

Use your point estimate and standard error to create and interpret a 95% confidence interval for the parameter.

$$CI_{\text{upper}} = 11.76 + 2 * 3.08 = 17.92$$

$$CI_{\text{lower}} = 11.76 - 2 * 3.08 = 5.6$$

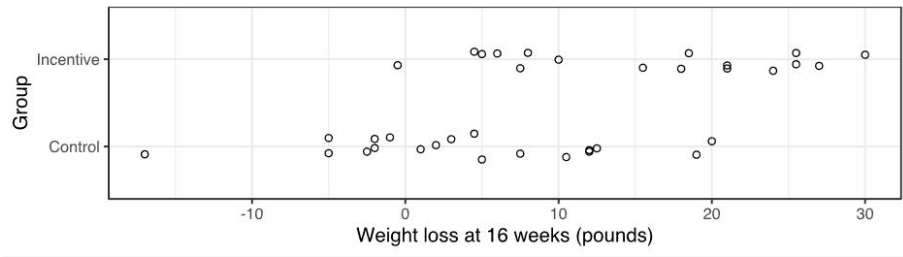
$$95_CI = (5.6, 17.92)$$

95_CI: we are 95% confident that people in the incentive group are between 5.6 and 17.92 pounds more than they would in the control group.

Group 10

The table and graph below show the actual experimental data from the weight-loss experiment that we studied earlier in the semester. Remember, the incentive group was given a financial incentive to lose weight.

Group	n	group mean weight loss at 16 weeks (pounds)	group std. dev. weight loss (pounds)
Control	19	3.92	9.11
Incentive	17	15.68	9.41



What parameter are we interested in estimating using the study data?

The difference in means between a control group and a money incentive to lose weight.

What is the “point estimate” (sample statistic) for this parameter?

$$15.68 - 3.92 = 11.76$$

What is the estimated standard error for the parameter. Include some of your work.

$$\text{Control: } 9.11 / \sqrt{19} = 2.09$$

$$\text{Incentive: } 9.41 / \sqrt{17} = 2.28$$

$$Se = \sqrt{2.09^2 + 2.28^2} = 3.092$$

Use your point estimate and standard error to create and interpret a 95% confidence interval for the parameter.

$$11.76 + 2 * 3.092 = 17.944$$

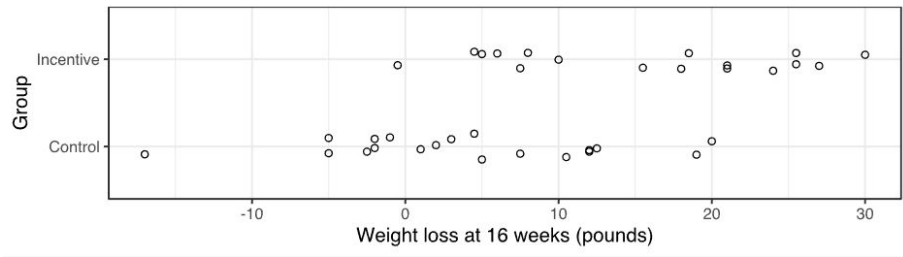
$$11.76 - 2 * 3.092 = 5.576$$

We are 95% confident that a person that if a person was in the incentive group they could lose anywhere between 5.576 pounds to 17.944 pounds more than they would have in the control group.

Group 11

The table and graph below show the actual experimental data from the weight-loss experiment that we studied earlier in the semester. Remember, the incentive group was given a financial incentive to lose weight.

Group	n	group mean weight loss at 16 weeks (pounds)	group std. dev. weight loss (pounds)
Control	19	3.92	9.11
Incentive	17	15.68	9.41



What parameter are we interested in estimating using the study data?
The difference in means between the control and incentive groups

What is the “point estimate” (sample statistic) for this parameter?

$$15.68 - 3.92 = 11.76$$

What is the estimated standard error for the parameter. Include some of your work.

$$9.11/\sqrt{19} = 2.08997733556$$

$$9.41/\sqrt{17} = 2.28226023159$$

$$\sqrt{(2.08997733556^2 + 2.28226023159^2)} = 3.09462712259$$

Use your point estimate and standard error to create and interpret a 95% confidence interval for the parameter.

$$11.76 \pm (1.96 * 3.09462712259) = 5.69453083972, 17.8254691603$$

We are 95% confidence that the mean difference will fall between 5.69453083972 and 17.8254691603